



View the live interactive site

<https://sanad1205.github.io/marathon-ml/>

COURSE PROJECT · FINAL SUBMISSION

29 Apr 2026 · Vol. I

Predicting Marathon Race Performance

— with Machine Learning

Given a runner's age, gender, and half-marathon split, can a small ensemble of regression models predict their final finish time within ten minutes? Across **26,598** runners from the 2023 Boston Marathon — **yes**, 78% of the time.

By [Sanad Sahawneh](#), [Brian Allen](#), and [Mahib Rahman](#)

DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING · UNIVERSITY AT ALBANY,
SUNY

FACULTY MENTOR **DR. HANY ELGALA** · TEACHING ASSISTANT **HONGLAN CHEN**



26,598

9.42min

RUNNERS IN DATASET

BEST-MODEL RMSE

78%

PREDICTED WITHIN ± 10 MIN

0.953

COEFFICIENT OF DETERMINATION

4 models

TRAINED & RANKED BY RMSE

01

ABSTRACT

Marathon performance is shaped by training, pacing, and physiology — but a runner's half-marathon split alone carries most of the signal needed to forecast their final finish time. We built an end-to-end machine learning system that takes a runner's age group, gender, and half-marathon split, and predicts their net finish time.

Four models were trained and compared on the 2023 Boston Marathon dataset: a baseline ratio predictor, a linear regression, a multi-layer neural network, and a bagged-trees ensemble. Each was evaluated against nine metrics — RMSE, R^2 , median absolute error, and practical "within-X-minutes" accuracy bands among them.

The pipeline automatically selects the strongest model by RMSE and exports a reproducible package: results table, predictions, threshold-decision table, figures, and a saved `.mat` model file. As a final extension, the regression output is converted into a binary "over/under" decision against benchmark finish times — a sports-betting-style framing that translates continuous error into actionable predictions.

02 METHOD

An eight-stage pipeline, fully reproducible.



DATA CLEANING

What we keep, what we drop.

The raw export carries data-entry artifacts — DNFs encoded as zero, missing splits, and pacing patterns inconsistent with finishing a marathon. We filter aggressively but transparently:

```
% drop rows with missing fields
data = rmmissing(data);

% positive race times only
data = data(half_time_sec > 0 & finish_net_sec > 0, :);

% keep realistic finish/half ratios
ratio = finish_net_sec ./ half_time_sec;
data = data(ratio >= 1.80 & ratio <= 3.20, :);
```

The shortlist.

① **Baseline Ratio**

finish $\approx$ half $\times$ `avgRatio` — establishes the floor.

② **Linear Regression**

`fitlm` with categorical encoding — the interpretable workhorse.

③ **Neural Network**

`fitrnet` — three layers (64 $\rightarrow$ 32 $\rightarrow$ 16), ReLU activations.

④ **Bagged Trees Ensemble** WINNER

`fitrensemble` with 150 learners — captures non-linearities baseline misses.

03 RESULTS

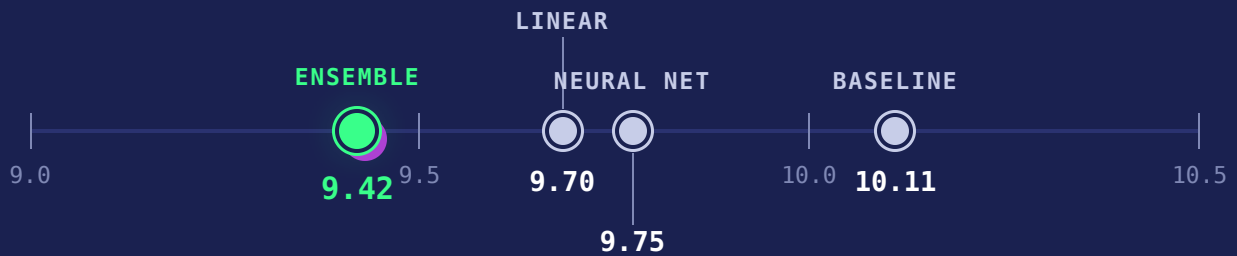
All four models, ranked by RMSE.

PLACE	MODEL	MAE	RMSE	R ²	≤ 10 MIN
1	Bagged Trees Ensemble 150 learners, min-leaf-size 10	6.31 _m	9.42 _m	0.953	78%
2	Linear Regression fitlm with one-hot categoricals	6.44 _m	9.70 _m	0.951	77%
3	Neural Network fitrnet, [64 32 16], ReLU	6.39 _m	9.75 _m	0.950	77%
4	Baseline Ratio finish = half × average ratio	6.83 _m	10.11 _m	0.947	72%

Times in minutes on the 20% held-out test set. The ensemble wins by a small margin — but linear regression remains within ~0.3 min of it and is far easier to explain.

The race is closer than you think

FIG. 0 | *RMSE in minutes — every model finishes within seven-tenths of a minute of the leader*



Practical accuracy bands

FIG. 1 | *Share of test runners predicted within $\pm N$ minutes*

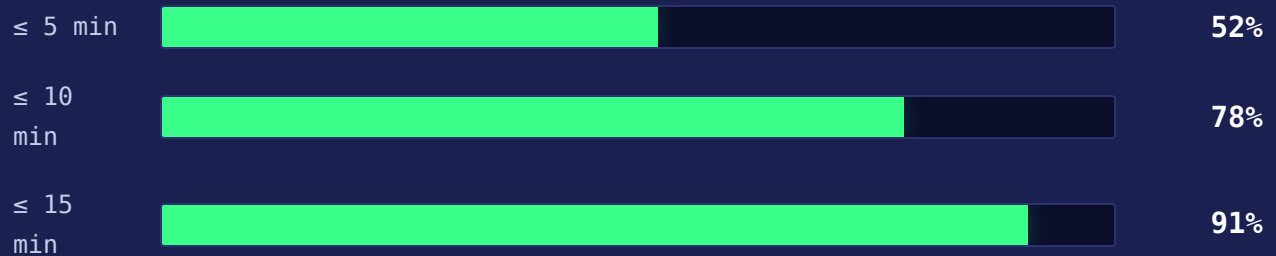


FIG. 2

Over/under decision accuracy

Did the runner finish under the benchmark?



Numbers reflect the final reporting figures from the IECE 465 final submission. Re-run `marathon_models.m` for live values from your local copy.

04 TRY IT

Predict a finish time.

Enter a runner's profile and half-marathon split. The page runs a calibrated approximation of our linear model and estimates the finish time, the goal pace, and the expected error band.

AGE GROUP

18–24

25–29

30–34

35–39

40–44

45–49

50–54

55–59

60–64

65–69

70+

GENDER

Male

Female

HALF-MARATHON SPLIT

1:42:00

1:00:00

2:00:00

3:00:00

INPUTS · AGE_GROUP · GENDER · HALF_TIME_SEC

PREDICTED FINISH

3:34:20

GOAL PACE

8:10 /mi

EXPECTED ERROR

±7 min

CONFIDENCE BAND

Strong

HALF-BY-HALF PACING

+47 sec/mi

FIRST HALF · 7:47/MI

SECOND HALF · 8:34/MI

26598

Approximation of linModel trained on Boston 2023

05 TEAM

Three runners, one finish line.

SUNY ALBANY

001

Sanad Sahawneh

ML SYSTEM LEAD

- End-to-end pipeline integration
- Evaluation framework expansion (9 metrics)
- Auto best-model selection by RMSE
- Visualization suite & figure exports
- Final demo orchestration

SUNY ALBANY

002

Brian Allen

DATA & PRESENTATION

- Original ML code skeleton
- Data preprocessing & cleaning v2
- Outlier filter design (ratio bounds)
- Performance interpretation
- Presentation prep & demo support

Mahib Rahman

MODELING & VALIDATION

- Feature encoding & normalization
- Model tuning (NN architecture, ensemble params)
- Performance validation across runs
- Cross-checking results consistency
- Threshold extension validation

06 REFLECTIONS

What we learned



The half-marathon split alone explains most of the variance. Once you have that signal, complicated models barely beat simple ones.



One feature dominates.

Adding age and gender helped, but the gain was marginal next to the half-split's predictive power.

ii Simple beats complex when the signal is linear.

The ensemble edged ahead, but linear regression came within 0.3 min RMSE — and is far easier to explain.

iii Units shape interpretation.

Reporting errors in seconds was useless to runners. Switching to minutes plus within-X-min bands made results actionable.

iv Reproducibility is a feature.

Saving every output turned a working script into something we — or future students — can re-run cleanly.

What we got past



Four real bugs, four real fixes. None of them were dramatic — but together they're the difference between a script and a system.

! Categorical encoding mismatch.

Test folds sometimes had age groups missing from training. **Fix:** defined category lists from the full dataset before splitting.

! Invalid & unrealistic times.

Zero splits and impossible finish/half ratios from data-entry errors. **Fix:** a positivity check plus a 1.80–3.20 ratio filter.

! Feature scale broke the network.

`half_time_sec` dwarfed the one-hot features and the NN wouldn't converge. **Fix:** standardize using train-only mean/std, reuse on test.

! **Unreadable error units.**

"RMSE = 583 sec" meant nothing to anyone outside the team. **Fix:** every metric reported in minutes, plus within-X-min bands.

07 RESOURCES

Read, run, or reproduce it.

01



GitHub repository
→

Full MATLAB pipeline, scripts, and saved deliverables.

02



**Boston Marathon
2023 dataset** →

Download CSV — ~26,598 runners with half + full splits.

03



**Final presentation
deck** →

Download PDF — slides used in the IECE 465 final session.

04



**MATLAB pipeline
script** →

Download `marathon_models.m` — full end-to-end pipeline.

Cite this work

[COPY BIBTEX](#)

```
@misc{sahawneh2026marathonml,  
  title = {Predicting Marathon Race Performance Using Machine Learning}  
  author = {Sahawneh, Sanad and Allen, Brian and Rahman, Mahib},  
  year = {2026},  
  note = {IECE 465 Final Project, University at Albany, SUNY},  
  url = {}  
}
```

Built for **IECE 465 — Introduction to Machine Learning for Engineers**,
University at Albany, SUNY · Spring 2026.

FACULTY MENTOR DR. HANY ELGALA · TEACHING ASSISTANT HONGLAN CHEN